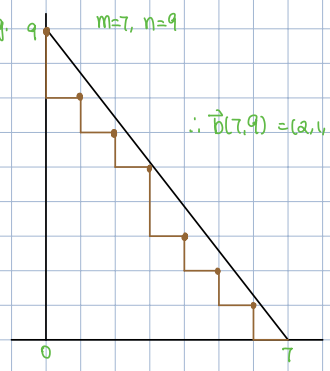
For $(m,n) \in \mathbb{Z}_+ \times \mathbb{Z}$ (not necessarily coprime), define a vector of integers $\vec{b}(m,n)$ of length m by

$$b(m,n)_i := \left\lceil i\frac{n}{m} \right\rceil - \left\lceil (i-1)\frac{n}{m} \right\rceil \qquad (i = 1, \dots, m).$$

* If $n \geq 0$, then $\vec{b}(m,n)_i$ = # S-steps on the line $x = i-1$ in the highest SE lattice path weakly below the line segment from $(0,n)$ to $(m,0)$.

e.g. $m=7$, $n=9$



$\therefore \vec{b}(7,9) = (2,1,1,2,1,1,1)$

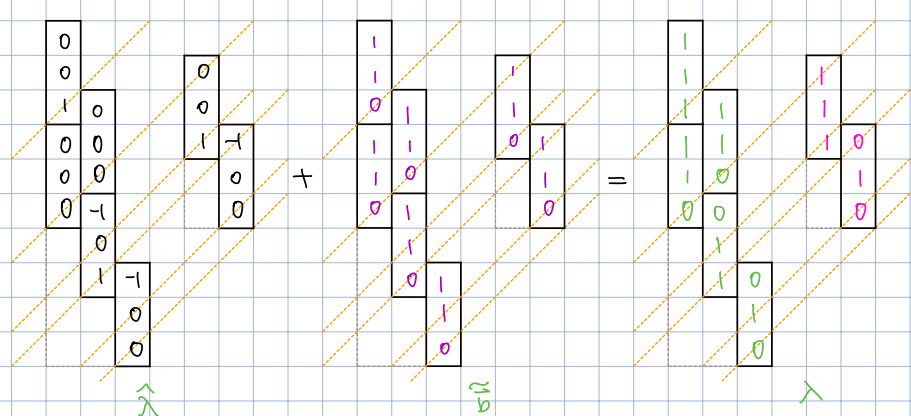
* $\vec{b}(dm,dn)$ is just the concatenation of d copies of $\vec{b}(m,n)$.

Def 8.2.1: Let
- $(m,n) \in \mathbb{Z}_+ \times \mathbb{Z}$ be a pair of coprime integers.
- ν be a tuple of skew shapes with $d = |\nu|$ boxes.
- $\nu^m = \nu(m,0)$ be an m-stretching of ν.
- $\ell = |\nu^m|$ ($\therefore \ell = dm$)

Define the LLT Catalanimal $H^{m,n}_{\nu^m} = H(R_q, R_t, R_{qt}, \lambda)$ as follows:
- R_q, R_t, R_{qt} as usual with ν replaced by ν^m
- $\lambda = \hat{\lambda} + \tilde{b}$ where
  - $\hat{\lambda}$ as usual with ν replaced by ν^m
  - $\tilde{b} \in \mathbb{Z}^\ell$ with $\tilde{b}_i = \vec{b}(m,n)_{\underbrace{\mathrm{mod}_m(c-o_s)}_{\in [m] \text{ equivalent to } c-o_s \pmod m}}$, c = content of $\nu^m(i) \in \nu^{m,o_s}_{(s)}$

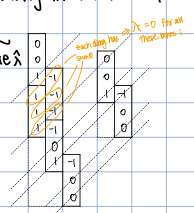
* λ is obtained from $\hat{\lambda}$ by adding the vector $\vec{b}(m,n)$ to each $\underbrace{\text{stretched box}}_{\text{set of } m \text{ boxes in } \nu^m}$ from north to south (see example below)

e.g. $\nu = (3^3_0, 2^2_0)$, $m=3$, $\vec{o} = (-4,-2)$ as in §8.1. Set $n=2$. Then $\vec{b}(3,2) = (\lceil \frac{2}{3} \rceil, \lceil \frac{4}{3} \rceil - \lceil \frac{2}{3} \rceil, \lceil \frac{6}{3} \rceil - \lceil \frac{4}{3} \rceil) = (1, 2-1, 2-2) = (1,1,0)$. To find λ:



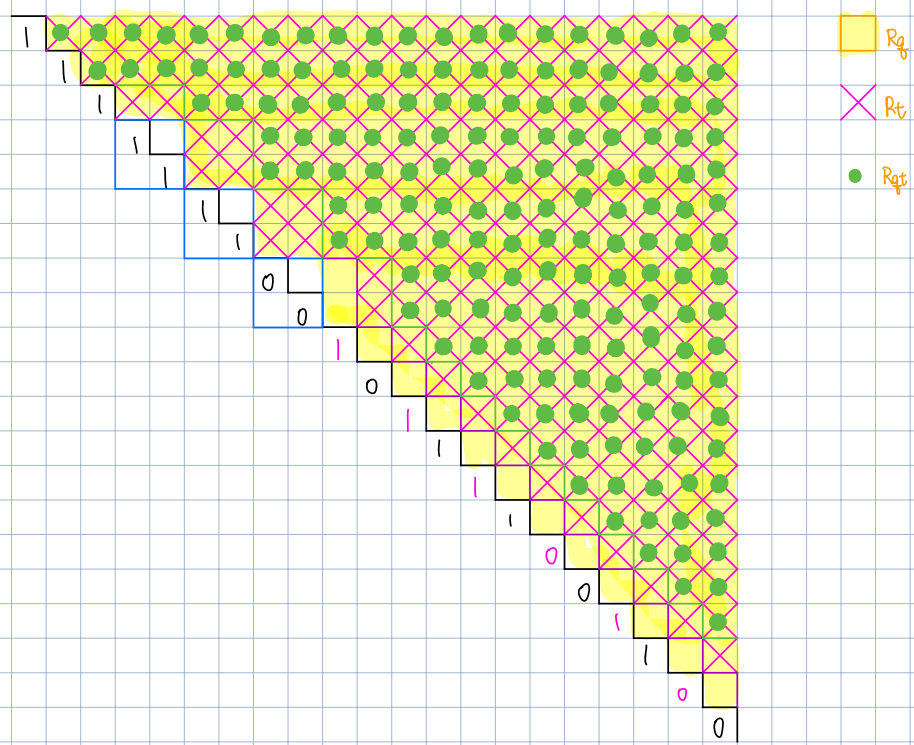
filling in box $i = \hat{\lambda}_i$

To compute $\hat{\lambda}$

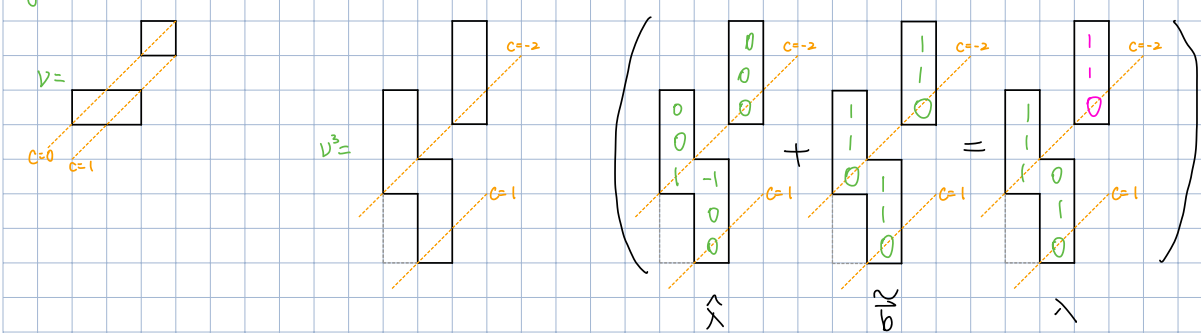


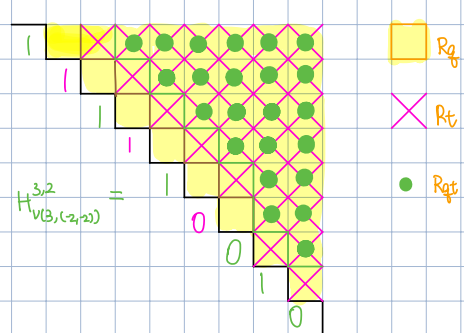
$\lambda = (1,1,1,1,1,1,1,1,0,0,1,0,1,1,1,1,0,0,1,1,0,0)$

The LLT Catalanimal $H^{3,2}_{\nu(3,(-4,-2))}$ for $\nu=(32/00, 2/0)$



e.g. $\nu=(2/0, 1/0)$, $m=3$, $n=2$, $\vec{b}=(-2,-2)$. Then $\vec{b}(3,2)=(1,1,0)$

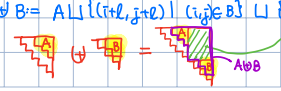




Remark: Define a binary operation $\uplus$ on sets of roots as follows:

For $A \subseteq R_+(GL_\ell)$ and $B \subseteq R_+(GL_{\ell'})$

$$A \uplus B := A \sqcup \{(i+\ell, j+\ell) \mid (i,j) \in B\} \sqcup \{(i,j) \mid 1 \le i \le \ell,\ \ell < j \le \ell+\ell'\} \subseteq R_+(GL_{\ell+\ell'})$$



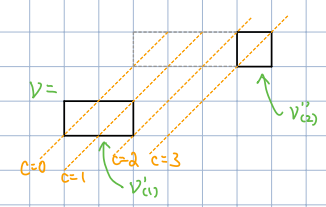
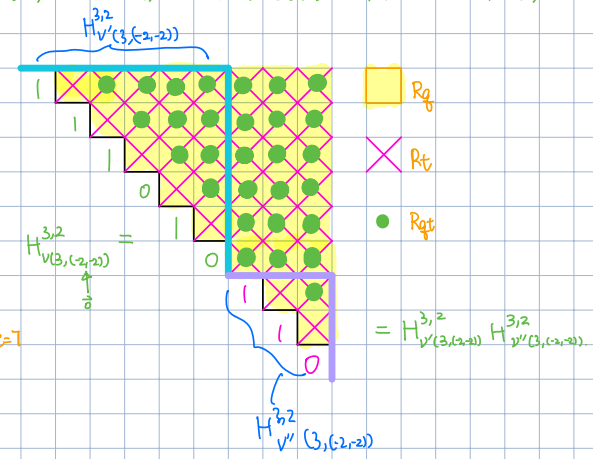
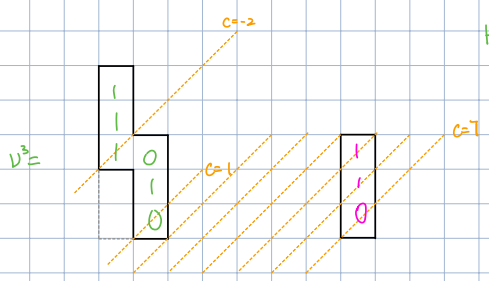
The product of two Catalanimals $H = H(R_q, R_t, R_{qt}, \lambda)$ and $H' = H(R_q', R_t', R_{qt}', \lambda')$ in $\mathcal{S}_A$ is another Catalanimal

$$HH' = H(R_q \uplus R_q',\ R_t \uplus R_t',\ R_{qt} \uplus R_{qt}',\ \underbrace{(\lambda;\lambda')}_{\text{just concatenation of } \lambda \text{ and } \lambda'})$$

The definition of LLT Catalanimals $H^{m,n}_{\nu(m,\vec{o})}$ interacts well with this product in the following sense:

$\nu_{(i)} = \nu'_{(i)} \sqcup \nu''_{(i)}$ If $\nu = \nu' \sqcup \nu''$ s.t. $\tilde{c}(a) + 1 < \tilde{c}(b)$ for all boxes $a \in \nu'$ and $b \in \nu''$, then the root sets and weights of $H^{m,n}_{\nu(m,\vec{o})}$ are constructed from those of $H^{m,n}_{\nu'(m,\vec{o})}$ and $H^{m,n}_{\nu''(m,\vec{o})}$, giving $H^{m,n}_{\nu'(m,\vec{o})} H^{m,n}_{\nu''(m,\vec{o})}$ in $\mathcal{S}_A$.

if the shapes are far apart enough

Hence one of $\nu'_{(i)}$ and $\nu''_{(i)}$ is empty for each i

e.g. $\nu = (2/0,\ 4/3)$, $m=3$, $n=2$, $\vec{o} = (-2,-2)$, $\vec{b}(3,2) = (1,1,0)$. $\nu' = (2/0,\ 0)$, $\nu'' = (0,\ 4/3)$